Repechage

Agentic Payment Recovery
for Failed Razorpay Checkouts

AI reasons. Policy decides. Execution is controlled.

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Razorpay AI Buildathon — Track 03

Agenda

From failed payment to controlled recovery

01
The Problem
Why revenue disappears

02
The Solution
How Repechage thinks

03
The System
Architecture & safety

04
The Evidence
Baseline vs. AI

We will end with what actually worked — and what didn’t.

Then the payment failed.

The opportunity existed. The recovery decision didn’t.

Payments fail for different reasons.

So why should they receive the same response?

`card_declined`

`insufficient_funds`

`otp_timeout`

`network_error`

Naive automation

Same response

The consequence

- Wasted recovery attempts
- Annoyed customers
- Missed recoverable payments
- No clear stopping point

Different failures need different decisions.

The opportunity is measurable.

₹98,24,112

total transaction value in our 1,000-event benchmark

66.6%

of revenue flagged at-risk

₹65,38,889

438

genuinely recoverable events

if the right action is taken quickly

This is not just a payment problem. It is a revenue recovery problem.

It knows what happened.

It doesn’t understand what to do next.

1

No diagnosis
Failure reasons are treated similarly.

2

No stopping rule
Unrecoverable payments keep getting contacted.

3

No audit trail
Merchants cannot easily explain automated actions.

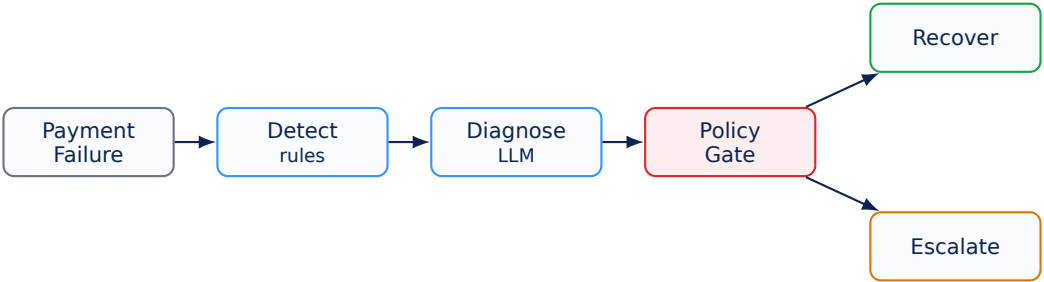
Detect. Diagnose. Recover.

An agent that reasons about the recovery decision — not just the failure.

But the AI does not get the final say.

AI reasons.
Policy decides.
Execution is controlled.

The recovery workflow



Every decision produces an audit trail.

The agent has a bounded action space

It cannot invent an action.

01	02	03	04	05
`recover_now`	`send_payment_link`	`wait_and_retry`	`escalate_to_merchant`	`stop`

Five actions. No free-form execution.

Safety lives in the policy layer

The AI recommends. Hard rules can still say NO.

Hard stops

- Attempts $\geq 3 \rightarrow$ **STOP**
- Already recovered $\rightarrow$ **STOP**
- Amount $>$ ₹5,000 $\rightarrow$ **ESCALATE**

Execution limits

- Maximum 10 real actions per run
- Live execution defaults to **OFF**
- Invalid or low-confidence outputs are gated

When in doubt: do not execute. Escalate instead.

Two models. Two jobs.

Deterministic Detector

```
status
failure_reason
previous_recovery_attempts
```

↓

Risk classification

Explainable · Zero LLM cost

LLM Recovery Agent

Rich transaction context

↓

Diagnosis + Action + Confidence

Contextual reasoning · Structured output

Neither model sees ground truth. Neither can bypass policy.

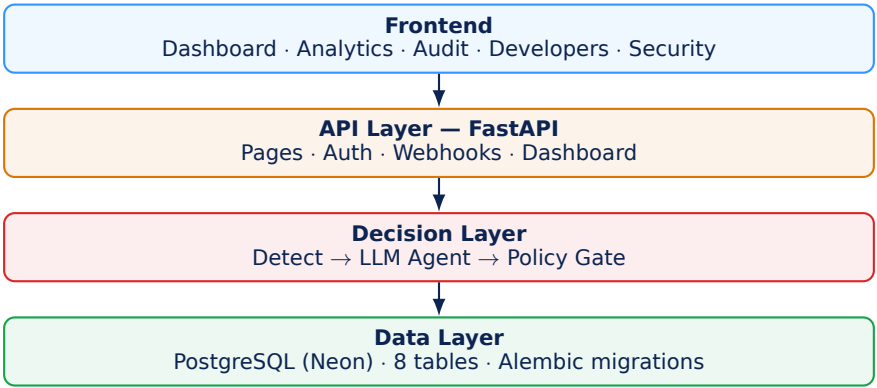
What does the AI actually return?

Not a paragraph. A structured decision.

```
{
  "diagnosis": "..."
  "recovery_probability": 0.82
  "recommended_action": "send_payment_link"
  "reason": "..."
  "confidence": 0.91
}
```

Schema validated → post-filtered → policy gated

System Architecture



Every decision leaves a trace

The system is designed to be explainable after the fact.

Transaction lifecycle

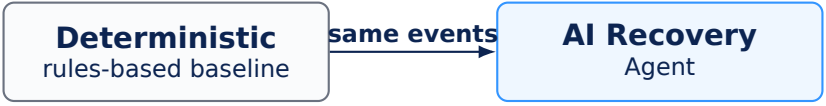
- `transactions`
- `synthetic_events`
- `detection_results`
- `agent_decisions`

Execution lifecycle

- `recovery_attempts`
- `audit_logs`
- `webhook_events`
- `merchants`

What happened? Why? When? What action was taken?
The audit layer keeps the answer.

Better than what?



Both systems are evaluated on the exact same 662 at-risk events under the same simulation economics.

The experiment

662

at-risk events evaluated

Deterministic Baseline	AI Recovery Agent
662 candidates	408 candidates
438 recoveries	298 recoveries
₹42,88,918 recovered	₹25,76,773 recovered
224 bad interventions	110 bad interventions

The AI was more precise.

73.0%
AI hit-rate

66.2%
baseline hit-rate

But it acted on far fewer opportunities.

408 vs. 662 candidates

Precision alone is not the objective.

Precision ↑

Volume ↓

Net revenue ↓

The next version must recover more of the right opportunities
without sacrificing the safety improvements.

That is the engineering lesson from the experiment.

A failed payment is not necessarily lost revenue.

AI reasons. Policy decides. Execution is controlled.

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Thank you.